Signature Verification Report

Professional biometric analysis — 1 sample(s) processed

March 06, 2026 at 06:38:57 UTC | bbc6a5f2c234... | Hybrid Engine v3 | 1 sample(s)

1 Executive Summary

SCOPE

This session processed
1 QUERY SIGNATURE SAMPLE(S)
against a pre-enrolled reference profile using HandAuth Pro's hybrid verification pipeline (deep metric learning + classical computer vision). The report covers all per-sample metrics, forensic indicators, and an overall session verdict.

OVERALL VERDICT — ✅ LOW RISK

Out of
1
sample(s):
1 LIKELY GENUINE (100.0%)
/
0 LIKELY FORGED (0.0%)
. Average calibrated confidence:
98.0%
. Presentation-attack alerts:
1
.

⚠ IMPORTANT LIMITATION

This automated assessment supports triage workflows only and is *not* a replacement for expert forensic document examination. For critical or legal decisions, qualified human review is mandatory.

2 Session Statistics

TOTAL SAMPLES	LIKELY GENUINE
1	1
Query signatures	100.0% of total

LIKELY FORGED	AVG CONFIDENCE
0	98.0%
0.0% of total	Calibrated mean

SESSION CONFIDENCE BREAKDOWN

CONFIDENCE DISTRIBUTION

0% 50% 100%

98.0% — ✅ LOW RISK

Values ≥70% = Likely Genuine | 50–69% = Uncertain | <50% = Likely Forged

KEY SESSION METRICS (FIRST SAMPLE)

Deep Cosine Similarity	0.9987
SSIM Index	0.845
Mahalanobis Distance	2.34
ORB Keypoint Matches	23

3 Per-Sample Detailed Assessment

HOW TO READ THESE CARDS

Each card covers one query sample. The score (top-right) is the calibrated probability of genuineness. Metric bars are normalised to their practical range — green = good, amber = borderline, red = poor.

1 Query #1: CamScanner 29.01.2026 14.04.pdf

Hybrid Analysis (Deep + Classical)

98.0%

✅ LOW RISK — LIKELY GENUINE

📋 **FINDING**

The deep learning model and classical metrics are broadly in agreement. The signature exhibits structural characteristics consistent with the enrolled reference. No significant anomalies were detected in edge-level or frequency-domain features.

METRIC BREAKDOWN

METRIC	VALUE	VISUAL	INTERPRETATION
Deep Cosine Similarity	**0.9987**		Very high embedding similarity — neural network considers signatures nearly identical.
SSIM (Structural Similarity)	**0.845**		Strong structural similarity — image regions closely match.
Mahalanobis Distance	**0.000**		Lower values indicate the query is closer to the enrolled reference cluster.
Pixel Correlation	**0.000**		Values near 1.0 indicate high pixel-level similarity after spatial alignment.
ORB Keypoint Matches	**23**		Number of stable local feature descriptor matches found between images.
Presentation Attack Prob.	**0.0%**		Estimated likelihood of forgery / reproduction by forensic heuristics.

GENUINE CONFIDENCE BAND

0% 50% (threshold) 70% (genuine) 100%

98.0% — ✅ LOW RISK — LIKELY GENUINE

🔴 **Presentation Attack Detected:** The forensic sub-system flagged this sample as a potential forgery, reproduction, or digital manipulation. Manual review is mandatory.

4 Verification Results Summary

#	SAMPLE NAME	CONFIDENCE	CLASSIFICATION	PA DETECTED
1	CamScanner 29.01.2026 14.04.pdf	**98.00%**	✅ Genuine	Yes

5 Analysis Methodology

1 Image Pre-Processing

Each input is normalised: background suppressed via adaptive thresholding, converted to greyscale, CLAHE contrast enhancement applied. PDFs are rendered at 150 DPI (PyMuPDF / Pillow fallback) then Lanczos-resampled to the canonical input size (224×224 or 128×128 depending on backbone).

2 Deep Metric Embedding

Image passed through a fine-tuned ResNet-18 / EfficientNet backbone (triplet / contrastive loss) producing a unit-normalised L2 vector of dim 512. Degenerate outputs (all-zero / NaN) trigger automatic fallback to a secondary CNN embedder.

3 Classical Image Comparison

- **SSIM** — luminance, contrast & structural correlation (0–1).
- **Pixel Correlation** — Pearson coefficient on aligned intensities (−1 to 1).
- **ORB Matching** — stable keypoint descriptor matches after ratio test.
- **Mahalanobis Distance** — embedding-space distance normalised by covariance.

4 Score Fusion & Calibration

Weighted fusion of deep cosine + classical metrics → isotonic regression calibration (logistic sigmoid fallback). Identity override applied when deep cosine ≥ 0.95 *and* SSIM ≥ 0.90. Penalty factor applied when deep is high but classical metrics disagree.

5 Presentation Attack Detection

Forensic sub-system analyses channel correlations, edge standard deviation, high-frequency energy, and a binary CNN classifier to detect reproductions, copy-paste artifacts, or digital manipulation. PA probability > 0.50 triggers an explicit alert.

6 Decision Thresholds & Configuration

METRIC	THRESHOLD	ROLE
Deep Cosine Similarity	**≥ 0.90**	Primary neural similarity signal
SSIM Index	**≥ 0.65**	Structural spatial agreement
ORB Match Ratio	**≥ 0.15**	Local feature correspondence
Pixel Correlation	**≥ 0.20**	Pixel-domain consistency
Genuine (calibrated prob.)	**≥ 70%**	Likely Genuine classification
Uncertain band	**50–69%**	Inconclusive — manual review
Forged (calibrated prob.)	**< 50%**	Likely Forged classification
Consensus requirement	**≥ 2 metrics**	Multi-metric agreement for high confidence

7 Recommendations

Likely Genuine (≥ 70%)

Standard verification protocols apply. Retain this report in the audit trail. No additional action required unless contextual signals (metadata, timestamps) raise independent concerns.

Uncertain (50–69%)

Do not rely solely on this score. Collect additional reference samples, verify enrollment quality, and escalate to a qualified forensic document examiner before making a final determination.

Likely Forged (< 50%)

Treat as high-risk. Initiate a formal fraud investigation workflow. Do not accept this signature for authentication purposes without expert clearance.

Presentation Attack Alert

Obtain the original physical document for independent examination. Request a fresh specimen in a controlled environment. Do not process the submitted copy through standard approval workflows.

ℹ️ **General Best Practices**

Enrol ≥ 3–5 reference signatures under consistent conditions. Pre-convert PDF inputs to high-resolution PNG (≥ 300 DPI) for best accuracy. Combine automated scores with contextual signals (timestamp, IP, device fingerprint).

8 Glossary of Terms

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SSIM

Structural Similarity Index — perceptual metric measuring luminance, contrast and structure (0 = different, 1 = identical).

ORB

Oriented FAST and Rotated BRIEF — rotation-invariant keypoint detector and descriptor for local feature matching.

Cosine Similarity

Cosine of the angle between two embedding vectors (−1 to 1; near 1 = highly similar).

Mahalanobis Distance

Scale-invariant distance accounting for embedding covariance structure. Lower = closer to reference cluster.

Pixel Correlation

Pearson coefficient between aligned image pixel intensities (−1 to 1).

Calibrated Probability

Raw score transformed by isotonic regression / logistic calibration into a genuine-authorship probability (0–1).

PA / Presentation Attack

Attempt to spoof the system using a reproduction — photocopy, photograph, or digitally generated facsimile.

Embedding Vector

Compact numerical representation of an image from a neural network's penultimate layer.

Triplet Loss

Training objective pulling same-writer embeddings together and pushing different-writer embeddings apart.

CLAHE

Contrast Limited Adaptive Histogram Equalisation — enhances local contrast while suppressing noise amplification.

Isotonic Regression

Non-parametric monotone calibration method mapping raw scores to probabilities while preserving ordering.

Writer-Dependent / Independent

Dependent: model fine-tuned on enrolled individual. Independent: general model trained across many writers.

Digital & Cryptographic Signature Verification

Results of PAdES (PDF embedded digital signature), CAdES (CMS detached signature), and PDF structural analysis performed on the submitted document.

Check	Status	Details
Document comparison	**Differs**	Hash match: ❌ No · Similarity: 100.0% · Page count match: Yes
└ Difference	—	title: 'CamScanner 29.01.2026 13.58' vs 'CamScanner 29.01.2026 14.04'
└ Difference	—	subject: 'CamScanner 29.01.2026 13.58' vs 'CamScanner 29.01.2026 14.04'

9 Legal Disclaimer & Limitations

This report is produced automatically by **HandAuth Pro** and is intended for informational and forensic triage purposes only. It does not constitute a legal opinion, a certified forensic examination, or expert witness testimony.

Automated systems are subject to known limitations including natural intra-writer signature variation, image acquisition quality, compression artifacts, scanner distortions, and adversarial inputs. No automated system achieves 100% accuracy.

For legally binding determinations, results must be reviewed, confirmed, or refuted by a qualified forensic document examiner. The operators of this system accept no liability for decisions made solely on the basis of this automated report.

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